

Yashab Yadav

PhD Scholar at Department of Physics, Lehigh University

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Education

Lehigh University , Doctor of Philosophy	Aug '23 – Present
• Major: Physics; passed dissertation proposal and entered PhD candidacy (2026)	
Indian Institute of Technology, Jodhpur , Masters of Science	Aug '21 – May '23
• Major: Physics	
• CGPA: 7.5/10	
University of Delhi , Bachelor's of Science	Aug '18 – May '21
• Major: Physics	
• CGPA: 8.662/10	

Research Experience

Force production and network structure of dendritic actin networks (simulation)	Jun '24 – Present
• Using 3D Langevin dynamics simulation of a dendritic actin network on a micrometer-scale bead with explicit Arp2/3-mediated branching & linear nucleation, barbed-end capping, and ADF/cofilin-driven filament turnover.	
• Mapped the branching-capping rate phase space for symmetry breaking and sustained bead propulsion; investigated how linear filament nucleation rate affects network architecture and motility.	
Intracellular transport dynamics of actin in crawling cells (data analysis)	Oct '25 – Present
• Built a Python pipeline to analyze single-molecule speckle microscopy (SiMS) trajectory data of oligomeric actin in the lamellipodia of crawling cells	
• Displacement distributions along the retrograde flow axis are decomposed into a double-Gaussian model to separate freely diffusing from transiently network-bound actin populations.	
Study of Branched Flow of light and its control	Nov '22 – May '23
• Studied branched flow of light as observed in thin soap films via simulations.	
• Applied beam propagation method to quantify simulation parameters for the sustenance of branches produced by a Gaussian light source passing through a disordered lattice.	

Presentations

Biophysical Society Annual Meeting , San Francisco, CA	Feb '26
• Poster: <i>Force and network structure of branched actin networks</i>	

Teaching Experience

Teaching assistant for Introductory Physics Course , Lehigh University	Aug '24 – Dec '25
• Helping undergraduate students to review fundamental physics concepts.	
• Demonstrating how to approach/solve fundamental physics problems.	
Teaching assistant for Introductory Physics Lab , Lehigh University	Aug '23 – May '24
• Instructed undergraduate students in procedures to carry out basic experiments in physics.	

Scholarships & Achievements

INSPIRE: Scholarship for Higher Education	
• Scholarship by Department of Science & Education, India.	
• Offered to 12000 meritorious students in the age group 17-22 years, annually.	
Joint Entrance Exam (JAM)	
• Got all India rank of 876 on JAM, a competitive entrance exam for admission to masters program in top research institutes in India.	

Computer Skills

Languages: Python (NumPy, SciPy, Matplotlib, Pandas), C++, Java, $\text{\LaTeX}$
Technologies: Git, Jupyter, OVITO, Docker

Coursework

Soft Matter Physics, Introduction to Computational Physics, Classical Dynamics, Statistical Mechanics, Electromagnetism, Quantum Mechanics, Vector Calculus, Linear Algebra, Machine Learning for Physicists